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The Flight Control of Micro Quad-Rotor UAV Based on PID

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***Abstract*—Attitude estimation is the key part of the flight control system of micro quad-rotor UAV. To begin with, quaternion and complementary filter Attitude algorithm is designed to reduce the operation quantity of attitude estimation. And then using the PID control algorithm work out the attitude control. Finally, use the android mobile phone via Bluetooth control the flight status of UAV, The simulation results demonstrate that the PID control system is well to maintain stable flight of the aircraft.**

Keywords—micro quad-rotor UAV; flight control; quaternion; PID controls; android mobile phone

I. INTRODUCTION

At present, the micro quad-rotor UAV because of its simple structure, low cost and easy to maintain, it has been found a wide range of application in many fields like military, environment hazards monitoring and scientific research [1]. Attitude estimation is the key part of the UAV control system, attitude measurement system by reading the value of the attitude sensor, through the appropriate attitude algorithm method based on quaternion and complementary filter, whereby get the roll angle, pitch angle and yaw angle of the UAV. Although the use of a single micro-mechanical gyroscopes can measure the attitude angles, but its accuracy is difficult to be assured, so that only the gyroscope in combination with other attitude sensors can measure the long-term, reliable attitude angle. Based on the quaternion complementary filter attitude algorithm methods described here can be used effectively integrate the error of gyroscope and accelerometer, finally ensure the accuracy of attitude solution.

II. SYSTEM HARDWARE OF STRUCTURE

The hardware structure of quad-rotor UAV flight control system is shown in Fig. 1, including the micro quad-rotor UAV, ground communication switching system, and Android smart mobile：

Quad-rotor UAV includes power module, STM32 MCU, attitude sensors MPU6050, ultrasonic sensor US-100, wireless transceiver module NRF24L01 +, etc.;

Ground communication switch terminal includes STM32 control board, a wireless transceiver end NRF24L01 + and Bluetooth module. Bluetooth and network enabled android mobile phone.



Fig. 1. System hardware structure.

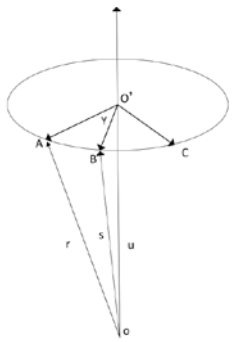


Fig. 2. UAV rotational movement.

III. ATTITUDE ESTIMATION OF MICRO QUAD-ROTOR UAV

A. Quaternions and Euler Angles

Suppose UAV airframe coordinate axes b is get by the geographic coordinate system g rotating around its three axes which according to the order of Z-axis, Y-axis, X-axis, and the rotation angle are, in order, α, β, θ (corresponding to roll angle, pitch angle and yaw angle) [2].

Quaternion Q is defined as:

$$Q(q_0, q_1, q_2, q_3) = q_0 + q_1 i + q_2 j + q_3 k \tag{1}$$

Where q_1, q_2, q_3 are real numbers, i, j, k are unit vector orthogonal to each other.

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Take a point on the UAV to analysis, as shown in Fig. 2.

Where the point A rotation around point O, OA that is the initial position vector, after the elapsed time t which rotation angle is γ. A and B are in the same plane. Projection vector of u in the geographic coordinate system is $\mathrm{u}^g = \begin{bmatrix} l & m & n \end{bmatrix}^T$.

The transfer matrix between coordinate axes b to g can be depicted as follows

$$C_b^g = \begin{bmatrix} 1-2\sin^2\frac{\gamma}{2}(m^2+n^2) & -2n\sin\frac{\gamma}{2}\cos\frac{\gamma}{2}+2lm\sin^2\frac{\gamma}{2} & 2m\sin\frac{\gamma}{2}\cos\frac{\gamma}{2}+2ln\sin^2\frac{\gamma}{2} \\ 2n\sin\frac{\gamma}{2}\cos\frac{\gamma}{2}+2lm\sin^2\frac{\gamma}{2} & 1-(l^2+n^2)\sin^2\frac{\gamma}{2} & -2l\sin\frac{\gamma}{2}\cos\frac{\gamma}{2}+2mn\sin^2\frac{\gamma}{2} \\ -2m\sin\frac{\gamma}{2}\cos\frac{\gamma}{2}+2ln\sin^2\frac{\gamma}{2} & 2l\sin\frac{\gamma}{2}\cos\frac{\gamma}{2}+2mn\sin^2\frac{\gamma}{2} & 1-(m^2+l^2)\sin^2\frac{\gamma}{2} \end{bmatrix}. \quad (2)$$

Four elements of quaternion Q can be written as

$$\begin{bmatrix} q_0 & q_1 & q_2 & q_3 \end{bmatrix}^T = \begin{bmatrix} \cos\frac{\gamma}{2} & l\sin\frac{\gamma}{2} & m\sin\frac{\gamma}{2} & n\sin\frac{\gamma}{2} \end{bmatrix}^T \quad (3)$$

Then

$$C_b^g = \begin{bmatrix} 1-2(q_2^2+q_3^2) & 2(q_1q_2-q_0q_3) & 2(q_1q_3+q_0q_2) \\ 2(q_1q_2+q_0q_3) & 1-2(q_1^2+q_3^2) & 2(q_2q_3-q_0q_1) \\ 2(q_1q_3-q_0q_2) & 2(q_2q_3+q_0q_1) & 1-2(q_1^2+q_2^2) \end{bmatrix} \quad (4)$$

Where

$$|Q| = q_0^2+q_1^2+q_2^2+q_3^2 = \cos^2\frac{\gamma}{2}+(l^2+m^2+n^2)\sin^2\frac{\gamma}{2} = 1 \quad (5)$$

Coordinate transformation matrix can be represented by

$$C_b^g = \begin{bmatrix} \cos\alpha\cos\beta & -\sin\alpha\cos\theta+\cos\alpha\sin\beta\sin\theta & \sin\alpha\sin\theta+\cos\alpha\sin\beta\cos\theta \\ \sin\alpha\cos\beta & \cos\alpha\cos\theta+\sin\alpha\sin\beta\sin\theta & -\cos\alpha\sin\theta+\sin\alpha\sin\beta\cos\theta \\ -\sin\beta & \cos\beta\sin\theta & \cos\beta\cos\theta \end{bmatrix}. \quad (6)$$

The relationship between the Euler angles and quaternion of quad-rotor UAV can be express as follows

$$\begin{bmatrix} \theta \\ \beta \\ \alpha \end{bmatrix} = \begin{bmatrix} \arctan\frac{2(q_2q_3+q_0q_1)}{1-2(q_1^2+q_2^2)} \\ \arcsin 2(q_0q_2-q_1q_3) \\ \arctan\frac{2(q_1q_2+q_0q_3)}{1-2(q_2^2+q_3^2)} \end{bmatrix}. \quad (7)$$

So, we need calculate the value of q_0, q_1, q_2, q_3 to get attitude angles like α, β, θ thereby controlling the movement of the UAV.

B. Attitude Estimation Algorithm

According to the definition of quaternion, Q can be depicted as follows:

$$Q = \cos\frac{\gamma}{2}+l\sin\frac{\gamma}{2}i+m\sin\frac{\gamma}{2}j+n\sin\frac{\gamma}{2}k = \cos\frac{\gamma}{2}+u^g\sin\frac{\gamma}{2} \quad (8)$$

$$\frac{dQ}{dt} = -\frac{\gamma}{2}\sin\frac{\gamma}{2}+u^g\frac{\gamma}{2}\cos\frac{\gamma}{2}+\sin\frac{\gamma}{2}\frac{du^g}{dt}. \quad (9)$$

According to the Coriolis theorem:

$$\frac{du^g}{dt} = C_b^g\frac{du^b}{dt}+\omega_{gb}^g\times u^g \quad (10)$$

Where u^g is projection vector of u in the UAV airframe coordinate system, ω_{gb}^g the rotation angle of the coordinate system b to the coordinate g under the geographic coordinate system g.

$$\frac{du^b}{dt} = 0, \omega_{gb}^g = \frac{d\gamma}{dt}u^g \quad (11)$$

So

$$\frac{du^g}{dt} = 0 \quad (12)$$

$$\frac{dQ}{dt} = -\frac{\gamma}{2}\sin\frac{\gamma}{2}+u^g\frac{\gamma}{2}\cos\frac{\gamma}{2} \quad (13)$$

$$\frac{d\gamma}{2dt}u^g\otimes Q = \frac{d\gamma}{2dt}u^g\cos\frac{\gamma}{2}-\frac{d\gamma}{2dt}\sin\frac{\gamma}{2} \quad (14)$$

$$\frac{dQ}{dt} = \frac{d\gamma}{2dt}u^g\otimes Q = \frac{1}{2}\omega_{gb}^g\otimes Q. \quad (15)$$

Since the angular velocity measured by attitude sensor is equal to the angular velocity of the body coordinate movement. So

$$\omega_{gb}^g = Q\otimes\omega_{gb}^b\otimes Q^* \quad (16)$$

Where ω_{gb}^b the rotation angle of the coordinate system b to the coordinates g under UAV airframe coordinate axes b.

$$\frac{dQ}{dt} = \frac{1}{2}\omega_{gb}^g\otimes Q = \frac{1}{2}Q\otimes\omega_{gb}^b\otimes Q^*\otimes Q = \frac{1}{2}Q\otimes\omega_{gb}^b. \quad (17)$$

Applying quaternion's multiplication formula, quaternions can be obtained as

$$\begin{bmatrix} \frac{dq_0}{dt} \\ \frac{dq_1}{dt} \\ \frac{dq_2}{dt} \\ \frac{dq_3}{dt} \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{bmatrix}\begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} \quad (18)$$

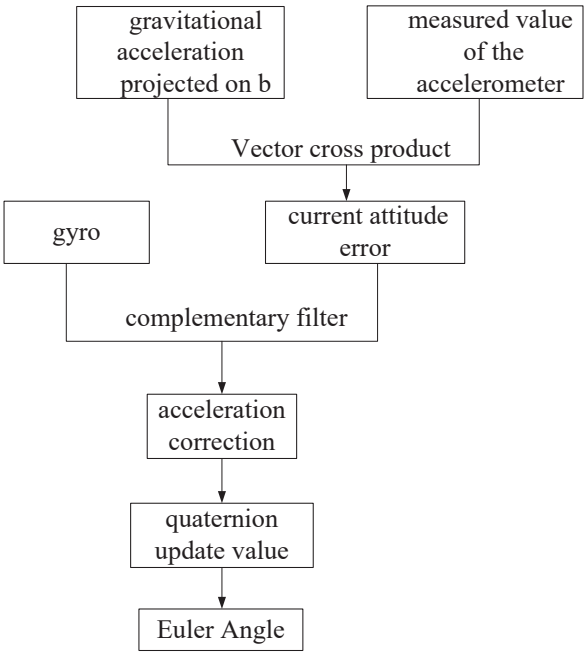


Fig. 3. Attitude estimation structure diagram.

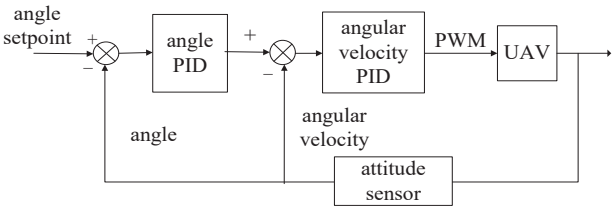


Fig. 4. PID control structure diagram.

where angular velocity ω_x, ω_y, ω_z are measured by attitude sensor, dq/dt is quaternion's increment in a short time, quaternion can obtained by integrating at a certain time, this time to calculate the UAV airframe Euler angles and then according to the formula (5), (6) can calculate UAV's attitude angle, thereby controlling the movement of the UAV. Attitude estimation structure diagram is shown in Fig. 3.

Due to the presence of integral calculation in attitude estimation, when angular velocity exist errors, may increases the integral error, it could lead to the obtained angle signal distortion. Then we need a good long-term performance of the accelerometer provide angle signal.

Through compare the actual acceleration get by three axis of body coordinates axes b to the acceleration projection on three axes of b, and then get the current attitude error to complementary filter with data obtained by gyro. Complementary filter coefficient is determined according to the actual need. Finally, we can obtain reliable Attitude angle [3].

C. *UAV Control Algorithm Design*

The system uses Android phone remote control the UAV's flight, like vertical take-off and landing, fly around, hover at a

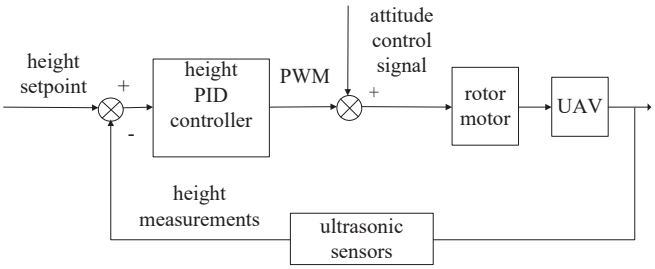


Fig. 5. Height control structure diagram.

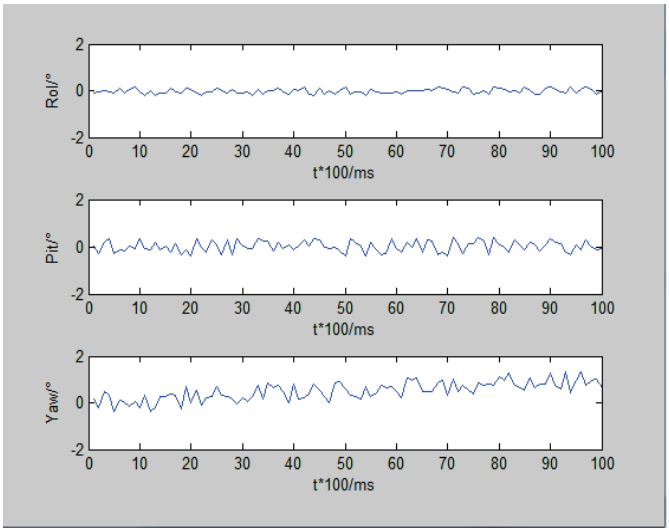


Fig. 6. Angular velocity simulation.

height. etc. Control algorithm mainly includes Double-loop PID algorithm, height control algorithm, digital filtering algorithm [4].

- Double-loop PID Control

Quad-rotor UAV's attitude control use Double-loop PID controller. Fig. 4 is a PID control structure diagram of quad-rotor UAV. Wherein the inner loop PID control loop once every 2ms, 5ms outer loop PID control gain.

- Height Control Algorithm

Fig. 5 is height control structure diagram of quad-rotor UAV.

- Digital Filter Algorithm

Digital filtering algorithms use first-order low-pass RC filter, small calculation applicability. When the input variable low frequency changes, this filtering algorithm is high efficient, can fully meet the control system requirements.

IV. Simulation Analysis

A. *Attitude angle Simulation*

In accordance with the principle of double-loop PID control, commissioning inner PID parameters firstly [5]. When setting the angular velocity target value as 0, simulation results is shown in Fig. 6.

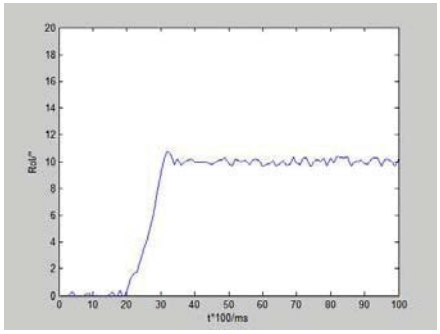
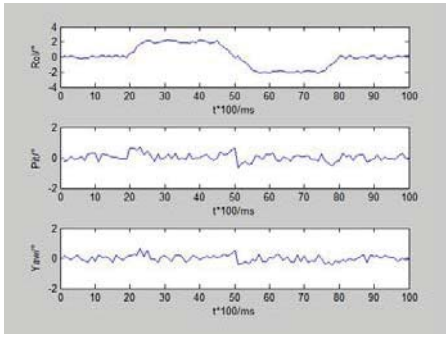
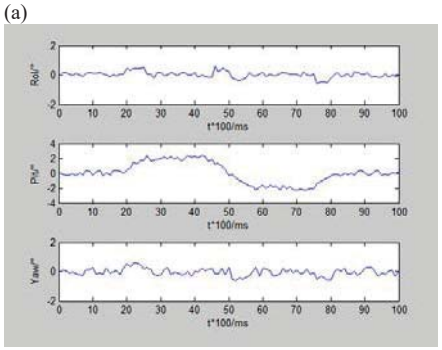


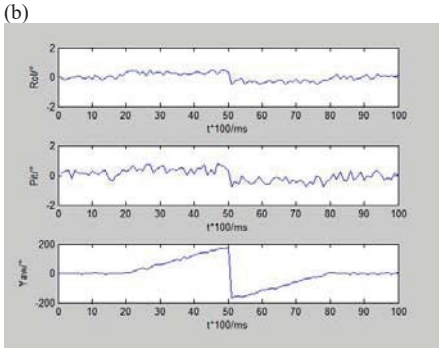
Fig. 7. Roll angle simulation.



(a)



(b)



(c)

Fig. 8. Flight simulation. (a) Horizontal flight angle simulation. (b) Horizontal flight angle simulation. (c) Rotary flight angle simulation.

After debugging inner loop PID parameters, then simulation outer PID parameters in every direction separately. Set the obtain angle as 10 °, three angles can quickly reach the target, meet the requirements and the steady-state error is small. Set roll angle as example, as shown in Fig. 7.

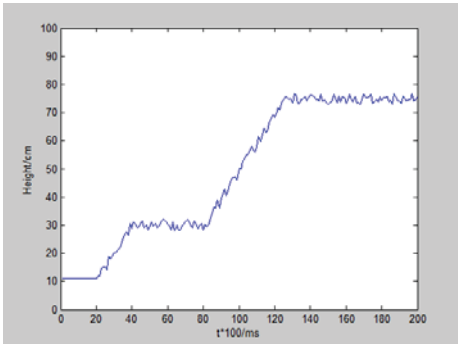


Fig. 9. PID control structure diagram.

B. Flight Test and Simulation

Using a mobile phone for control UAV flight in different directions, as shown in Fig. 8 shown the simulation of quad-rotor UAV flying around in horizontal (right and left), longitudinal (front and back) and rotary.

C. Height Test and Simulation

As shown in Fig. 9, the UAV's initial position at a height of 11cm, and then set the target height of 30cm and 75cm in order, the UAV reaches the specified height to maintain a balance by phone, held a height error at positive and negative 2cm, meet the control requirements.

CONCLUSION

Test and simulation demonstrate android mobile phone is able to achieve stable attitude control of micro quad-rotor UAV, completed the task and achieved the desired objectives. The introduction of android smart devices provides new ways to control the quad-rotor UAV, has a positive effect to practical application of micro four-rotor aircraft. Also, there are many parts need to be improvement in future studies, like design the more efficient system controller and development the mobile phone APP.

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